

Analysing Results from Molecular Docking

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1 Introduction

Molecular docking is now a routine component of structure-based drug discovery pipelines, used at scales ranging from hypothesis-driven pose analysis of a handful of ligands to ultra-large virtual screens of hundreds of millions of compounds. Despite this apparent maturity, the interpretation and validation of docking results remain nontrivial, and are often the weakest link in otherwise sophisticated workflows. Numerous benchmarking studies have shown that while many docking programs can often recover crystallographic binding modes with reasonable accuracy, their ability to rank ligands by affinity or to drive robust hit discovery is far more limited and system-dependent.[16, 8, 3]

This chapter focuses on practical aspects of analyzing and validating docking results. We emphasize not only the good practice that experienced practitioners follow, but also the common failure and negative examples that one encounters in the literature and in early-stage projects. The goal is to provide a realistic view of what docking can and cannot do, and how to design and interpret calculations so that they inform, rather than mislead.

2 Docking Outputs

A typical small-molecule docking calculation produces one or more putative binding modes, also called poses, for each ligand in a defined binding site. This is accompanied by a numerical score that is intended to correlate to binding affinity. More realistically, the score must rank ligands in a relative sense.[8, 10] Many modern workflows also provide additional data such as:

- Decomposition of interaction in to van der Waals, Coulombic, desolvation, ionic, and hydrogen bonding,
- interaction fingerprints encoding the residue interaction patterns

- pose clusters based on ligand heavy-atom RMSD,
- warnings about steric clashes or strained conformations.

The main challenge is that none of these quantities are experimentally observable. They are proxies for the thermodynamic and kinetic processes that determine drug binding. The main objective of any docking method relies on the ability of the search algorithm to find biologically reasonable poses. And the ability of the scoring function to rank those poses and ligands in a way that correlates with experimental affinity data.[15, 3]

3 Qualitative Analysis of Docking Poses

3.1 Visual Inspection

Despite extensive automation, visual inspection of docking poses remains indispensable. Most workflows almost always subject top-ranked poses to manual or semi-automated review by an experienced computational before making design or screening decisions.[8, 4]

Visual inspection should aim to assess:

- Geometric feasibility: Does the ligand sit sensibly in the pocket without severe steric clashes or burial of polar atoms in apolar environments without compensation?
- Key pharmacophoric interactions: Are canonical interactions, such as hinge hydrogen bonds in kinases, catalytic residues in proteases, metal-chelating motifs, satisfied?
- Internal strain: Are torsions in high energy conformations relative to known low energy conformers or crystallographic analogues?
- Water molecules: Are key waters displaced or retained in a manner consistent with known structures and thermodynamic expectations?

3.2 Negative Examples

Despite knowing what to expect while analysing docking outcomes, it is seldom discussed on what not accept while visualising docking poses. The following points are non-exhaustive recommendations:

- Buried, unsatisfied charges: Poses in which a formal charge, such as those from a protonated amine or deprotonated carboxylate group, is completely buried in a hydrophobic pocket without forming salt bridges or strong hydrogen bonds are artefacts, even if highly scored.
- Severe steric clashes: Ligand atoms overlapping protein atoms, or gross van der Waals clashes between rings and side chains, indicate a failure in the search algorithm or scoring functions. These poses should be discarded or the protocol re-examined.
- Unrealistic ligand contortions: Highly twisted ring systems or torsions that deviate dramatically from known crystallographic or QM-optimized conformations, without clear compensating interactions, suggest overfitting to the scoring function.
- Unphysical metal coordination: Poses that ignore established coordination geometries and preferred donor atoms for metal ions, like Zn^{2+} , Mg^{2+} , should be treated with great suspicion unless supported by strong experimental evidence.

4 Quantitative Analysis

Docking scores are often reported in units of binding free energy like kcal/mol or kJ/mol, however, they are rarely parameterised to yield quantitatively accurate ΔG values. Comparative assessments show that simple linear fitting between scores and experimental affinities are usually target-specific and often display large residuals.[16, 3] Therefore:

- treat scores as relative ranking tools within a given target and protocol,
- avoid comparing scores across different targets, protein conformations, or docking algorithms,
- be skeptical of fine distinctions in score, differences $< 1\text{--}2$ kcal/mol between ligands may well be within the error region of the scoring function, and not true difference in binding affinity.

4.1 Score Distributions and Rank-Based Analysis

Rather than focusing on individual scores, it is more robust to analyze:

- Score distributions: Histograms of scores for all docked ligands often reveal whether a few ligands are clear outliers or whether the distribution is narrow and undifferentiated.
- Rank-based enrichment: When known actives are present, one can compute enrichment factors (EF), receiver operating characteristic (ROC) curves, and early enrichment metrics such as BEDROC to evaluate how well the scoring function prioritizes actives near the top.
- Score vs. physicochemical properties: Plots of score versus molecular weight, cLogP, polar surface area, or charge can reveal systematic biases, such as those in the scoring functions that simply favor larger or more hydrophobic compounds.

4.2 Negative Examples

Problematic practices include:

1. Using raw scores as binding free energies: Statements such as the docking score of -11 kcal/mol indicates nanomolar affinity are not supported by benchmarking data and should be avoided.[16]
2. Hard score cutoffs without context: Applying an arbitrary score threshold, such as keep only ligands with score < -8 kcal/mol, without prior validation can remove true actives or retain large numbers of false positives.
3. Cross target score comparison: Comparing scores of the same ligand docked to two different proteins, and inferring selectivity directly from the difference is unjustified unless the scoring function has been carefully validated for that purpose.

5 Internal Validation of Docking Protocols

5.1 Redocking: Reproducing Known Binding Modes

Redocking tests use crystallographic complexes with known bound ligands. The ligand is removed, re-docked into the binding site, and the resulting poses are compared to the experimental binding mode, usually via RMSD. A typical acceptance criterion is that at least one of the top-ranked poses has heavy-atom RMSD < 2.0 Å from the crystal pose.[16, 8, 3]

Best practices for redocking include:

- using multiple co-crystal structures if available, to probe robustness across conformational states,
- evaluating both pose *reproduction* (RMSD) and *ranking* (whether the near-native pose is ranked among the top poses),
- ensuring that protein and ligand preparation protocols used in redocking match those planned for virtual screening.

A negative example is to report only the best RMSD achievable by any pose, while ignoring that the near-native pose was ranked very low; such protocols often fail in screening contexts.

5.2 Cross-Docking and Induced Fit Considerations

Cross-docking, in which ligands are docked into non-cognate protein conformations, assesses sensitivity to receptor flexibility and pocket variability. This is particularly relevant for highly flexible targets such as kinases, nuclear receptors, or GPCRs.[10, 9]

Key recommendations:

- include representative open and closed conformations when available,
- quantify success rates across multiple structures, not just a single “best” receptor,
- consider ensemble docking or induced-fit protocols when single-structure performance is poor.

A common mistake is to perform cross-docking on highly different conformational states (e.g., apo vs. holo with large loop movements) without adjusting the protocol or interpreting failures in light of structural differences.

5.3 Enrichment-Based Validation with Actives and Decoys

When a set of known actives is available, along with an appropriate set of inactive or decoy compounds, virtual screening-style validation can be performed.[14, 12]

Best practices for enrichment-based validation include:

- Property-matched decoys: Use decoy sets that match simple physico-chemical properties (e.g., molecular weight, cLogP, HBD/HBA counts) but differ in topology from actives; the DUD-E resource is a widely adopted standard.[12]
- Early recognition metrics: Compute enrichment factors at low percentages of the ranked list (e.g., EF_{1%}, EF_{2%}), ROC-AUC, and early recognition metrics such as BEDROC.[13]
- Multiple random splits: When decoys are custom-generated, use multiple random subsets to avoid overfitting to a single decoy realization.

Negative examples include:

1. using random “drug-like” compounds as decoys without property matching, leading to artificially inflated enrichment,
2. reporting only ROC-AUC without considering that high AUC can coexist with poor early enrichment, which is more relevant for experimental follow-up,[13]
3. validating on the same dataset used for method development, leading to optimistic performance estimates.

6 Retrospective and Prospective Benchmarks

6.1 Large-Scale Comparative Studies

Several large-scale comparative studies have systematically assessed docking programs and scoring functions across diverse targets, highlighting that:

- pose prediction is usually more reliable than affinity ranking,
- no single docking program or scoring function is universally superior,
- performance varies strongly by target class and binding site features.[16, 3, 15]

These studies motivate target-specific validation and, in many cases, the use of consensus strategies or post-docking rescoring.

6.2 Public Benchmark Sets

Public benchmarks such as DUD-E, DEKOIS, and others provide standardized actives and decoys across many targets, facilitating method comparison and protocol optimization.[12] However, users should be aware that benchmarks can still exhibit biases (e.g., analogue bias, decoy selection biases) and may not capture all nuances of a specific industrial target class.

7 Rescoring Strategies

7.1 Consensus Scoring

Consensus scoring combines outputs from multiple docking programs or scoring functions, typically by averaging scores, aggregating ranks, or using more sophisticated schemes such as weighted or exponential rank combinations.[3, 1]

Pros:

- can reduce method-specific noise and systematic biases,
- often improves enrichment in retrospective virtual screens,
- straightforward to apply when multiple methods are already in use.

Cons and pitfalls:

- requires careful standardization of input scores (e.g., sign conventions, scales),
- can be dominated by one poor-performing method if naive averaging is used,
- increases computational cost if additional docking runs are needed.

A negative example is to simply average raw, non-normalized scores from heterogeneous scoring functions without assessing their individual behavior, then claiming improvement based on a single test case.

7.2 Empirical and Knowledge-Based Rescoring Functions

Several stand-alone scoring functions (e.g., X-Score, DrugScore, PLP variants) were designed explicitly for rescoring of docked poses, often calibrated on curated protein–ligand complexes.[3] These can be applied to poses generated by any docking engine, decoupling pose generation from scoring.

Best practices:

- ensure that the parameterization domain of the rescoring function (e.g., target types, ligand chemotypes) overlaps with the intended application,
- validate rescoring performance using redocking and enrichment tests on relevant systems,
- examine correlation between original and rescored ranks to understand how much reordering occurs.

7.3 Physics-Based Rescoring: MM/PBSA and MM/GBSA

Physics-based post-processing methods such as MM/PBSA and MM/GBSA approximate binding free energies by combining molecular mechanics energies with continuum solvation models, often using snapshots from molecular dynamics (MD) simulations.[7, 6, 5] Multiple studies have shown that, when carefully applied, MM/GBSA can outperform standard docking scores in ranking congeneric ligands and in discriminating near-native from decoy poses.[7, 6, 2, 17]

Key considerations for MM/PBSA and MM/GBSA rescoring include:

- Sampling: Adequate conformational sampling is essential; short MD trajectories (1–5 ns) can suffice for many systems but may be inadequate for highly flexible binding sites.[7, 17]
- Force field and charge models: Choice of protein force field and ligand charge model (e.g., RESP vs. AM1-BCC) has a substantial impact on predictive performance, and best choices can be system-dependent.[17, 5]
- Entropy: Entropic contributions are often approximated or neglected; relative ranking within congeneric series can still be meaningful even without explicit entropy treatment, but absolute binding free energies may be less reliable.

Negative examples in MM/GBSA usage include:

1. reporting MM/GBSA values with four significant figures and interpreting them as precise ΔG values,
2. neglecting to specify key parameters (dielectric constants, GB model, number of snapshots, simulation length),
3. using MM/GBSA on single, unminimized docking poses without any relaxation, which can amplify artefacts from docking.

7.4 MD-Based Refinement

MD simulations can be used to relax docking poses and explore local conformational space, followed by rescoring with MM/GBSA or related schemes. Additionally, MD can reveal the stability of key hydrogen bonds, hydrophobic contacts, and water networks that are only crudely approximated in docking.[11, 6]

Practical tips:

- focus MD refinement on a manageable subset of promising ligands (e.g., tens to low hundreds),
- monitor RMSD of ligand and nearby residues to ensure that the complex remains physically reasonable,
- analyze occupancy of key waters, especially in cases where “displaceable” vs. “conserved” waters may influence potency.

8 Best-Practice Workflow in Industrial Docking Campaigns

8.1 Protein and Ligand Preparation

Robust protein and ligand preparation is foundational. Recommended steps include:

- choose high-quality experimental structures (resolution, B-factors, completeness), preferably in complex with ligands similar to the intended chemotypes;

- assign protonation states and tautomers for ionizable residues and ligands at relevant pH using established tools; consider alternative states for borderline cases;
- model missing loops and side chains where necessary, but avoid over-modelling when structural uncertainty is high;
- retain or remove crystallographic waters based on well-defined criteria (e.g., buried, conserved waters vs. bulk-like, weakly bound waters).

Negative examples:

1. using an apo structure with a collapsed binding pocket for docking without any attempt to open or relax it,
2. ignoring biologically relevant cofactors or metal ions (e.g., Mg^{2+} in kinases, Zn^{2+} in metalloproteases),
3. assuming all titratable groups are in their default protonation state at pH 7, regardless of microenvironment.

8.2 Docking Protocol Design

A typical industrial docking campaign might proceed as follows:

1. Protocol calibration: Perform redocking and enrichment-based tests on representative co-crystal ligands and in-house actives, adjusting parameters (grid size, sampling exhaustiveness, constraints) to achieve acceptable pose recovery and enrichment.
2. Primary screen: Dock a large library with high-throughput settings, using a validated scoring function and retaining multiple poses per ligand.
3. Post-processing: Apply property and substructure filters (e.g., removal of PAINS and undesirable reactive groups), cluster by chemotype, and perform qualitative pose inspection of top-ranked chemotypes.
4. Secondary scoring: Apply consensus scoring and/or MM/GBSA rescoring to a reduced set, followed by deeper visual curation.
5. Selection for synthesis or purchase: Prioritize compounds based on a combination of score, pose quality, novelty, synthetic tractability, and alignment with project hypotheses.

9 Common Pitfalls and Negative Examples

9.1 Score Worship and Over-Interpretation

One of the most pervasive errors is to treat docking scores as if they were experimentally measured affinities. Examples include:

- ranking analogues solely based on small score differences and proposing SAR trends without any validation,
- inferring potency categories (e.g., micromolar vs. nanomolar) directly from score ranges,
- claiming that a ligand “binds more strongly” than a reference compound purely because its docking score is more negative by 0.5–1.0 units.

Such claims are not supported by benchmarking data and often contradict observed uncertainties in scoring functions.[16, 3]

9.2 Ignoring Protonation, Tautomerism, and Stereochemistry

Another frequent failure mode is to dock a single arbitrary protonation state, tautomer, or stereoisomer of a ligand and then over-interpret the resulting pose and score. Negative examples include:

1. docking a ligand with multiple basic centers in a fully protonated form in a hydrophobic pocket,
2. ignoring alternative tautomers that would better complement the hydrogen bond pattern of the pocket,
3. neglecting to consider all stereoisomers when chirality is not experimentally defined.

Best practice is to enumerate chemically reasonable protonation states, tautomers, and stereoisomers, filter them based on pH and prior knowledge, and dock the relevant set, assessing whether the binding mode is robust across plausible states.

9.3 Cherry-Picking Poses and Ligands

Docking generates many poses and ligands; it is tempting to select only those that fit a desired hypothesis or structural narrative. Problematic behaviors include:

- highlighting a pose that matches a preconceived pharmacophore while ignoring that it is ranked very low or exhibits clashes,
- presenting only a few highly favorable scores while omitting the broader distribution and the presence of close analogues with poor scores,
- modifying the protein structure or protonation states post hoc to accommodate a particular ligand without systematically revalidating the protocol.

These practices bias interpretation and undermine the predictive value of the calculations.

9.4 Treating Docking as Proof of Binding

Docking is a hypothesis-generation tool, not proof of binding or mechanism. Negative examples include:

1. asserting that docking “confirms” experimental binding or mechanism solely because a plausible pose can be found,
2. using docking alone to claim that a compound is selective for one target over another,
3. drawing mechanistic conclusions (e.g., about covalent bond formation or proton transfer) from docked structures that do not include the necessary quantum-chemical detail.

A more appropriate stance is to treat docking as *supporting* evidence, to be weighed alongside experimental data and other computational methods.

9.5 Misuse of Decoy Sets and Benchmarks

Finally, validation itself can be misused. Examples include:

- tuning a scoring function or protocol on a benchmark set (e.g., DUD-E) and then claiming superior performance based on the same set, without external validation,[12]

- constructing decoy sets that are trivially separable from actives (e.g., by size or polarity), thereby inflating enrichment metrics,
- comparing methods using different decoy sets or evaluation metrics, leading to incommensurate results.

Best practice is to use independent training and test sets, employ property-matched decoys, and report multiple, complementary performance metrics.

10 Conclusions

To be written

References

- [1] Clara Blanes-Mira, Pilar Fernández-Aguado, Jorge de Andrés-López, Asia Fernández-Carvajal, Antonio Ferrer-Montiel, and Gregorio Fernández-Ballester. Comprehensive survey of consensus docking for high-throughput virtual screening. *Molecules*, 28(1):175, 2022.
- [2] Fu Chen, Hui Liu, Huiyong Sun, Peichen Pan, Youyong Li, Dan Li, and Tingjun Hou. Assessing the performance of the mm/pbsa and mm/gbsa methods. 6. capability to predict protein-protein binding free energies and re-rank binding poses generated by protein-protein docking. *Physical Chemistry Chemical Physics*, 18(32):22129–22139, 2016.
- [3] Tiejun Cheng, Xun Li, Yan Li, Zhihai Liu, and Renxiao Wang. Comparative assessment of scoring functions on a diverse test set. *Journal of chemical information and modeling*, 49(4):1079–1093, 2009.
- [4] Stefano Forli. Charting a path to success in virtual screening. *Molecules*, 20(10):18732–18758, 2015.
- [5] Samuel Genheden and Ulf Ryde. The mm/pbsa and mm/gbsa methods to estimate ligand-binding affinities. *Expert opinion on drug discovery*, 10(5):449–461, 2015.
- [6] Tingjun Hou, Junmei Wang, Youyong Li, and Wei Wang. Assessing the performance of the mm/pbsa and mm/gbsa methods: Ii. the accuracy of ranking poses generated from docking. *Journal of computational chemistry*, 32(5):866, 2010.

- [7] Tingjun Hou, Junmei Wang, Youyong Li, and Wei Wang. Assessing the performance of the mm/pbsa and mm/gbsa methods. 1. the accuracy of binding free energy calculations based on molecular dynamics simulations. *Journal of chemical information and modeling*, 51(1):69–82, 2011.
- [8] Douglas B Kitchen, Hélène Decornez, John R Furr, and Jürgen Bajorath. Docking and scoring in virtual screening for drug discovery: methods and applications. *Nature reviews Drug discovery*, 3(11):935–949, 2004.
- [9] Thuluz Meza Menchaca, Claudia Juárez-Portilla, and Rossana C Zepeda. Past, present, and future of molecular docking. In *Drug discovery and development-new advances*. IntechOpen, 2020.
- [10] Xuan-Yu Meng, Hong-Xing Zhang, Mihaly Mezei, and Meng Cui. Molecular docking: a powerful approach for structure-based drug discovery. *Current computer-aided drug design*, 7(2):146–157, 2011.
- [11] Xuan-Yu Meng, Hong-Xing Zhang, Mihaly Mezei, and Meng Cui. Molecular docking: a powerful approach for structure-based drug discovery. *Current computer-aided drug design*, 7(2):146–157, 2011.
- [12] Michael M Mysinger, Michael Carchia, John J Irwin, and Brian K Shoichet. Directory of useful decoys, enhanced (dud-e): better ligands and decoys for better benchmarking. *Journal of medicinal chemistry*, 55(14):6582–6594, 2012.
- [13] Jean-François Truchon and Christopher I Bayly. Evaluating virtual screening methods: good and bad metrics for the “early recognition” problem. *Journal of chemical information and modeling*, 47(2):488–508, 2007.
- [14] Marcel L Verdonk, Valerio Berdini, Michael J Hartshorn, Wijnand TM Mooij, Christopher W Murray, Richard D Taylor, and Paul Watson. Virtual screening using protein- ligand docking: avoiding artificial enrichment. *Journal of chemical information and computer sciences*, 44(3):793–806, 2004.
- [15] Renxiao Wang, Xueliang Fang, Yipin Lu, Chao-Yie Yang, and Shaomeng Wang. The pdbind database: methodologies and updates. *Journal of medicinal chemistry*, 48(12):4111–4119, 2005.

- [16] Gregory L Warren, C Webster Andrews, Anna-Maria Capelli, Brian Clarke, Judith LaLonde, Millard H Lambert, Mika Lindvall, Neysa Nevins, Simon F Semus, Stefan Senger, et al. A critical assessment of docking programs and scoring functions. *Journal of medicinal chemistry*, 49(20):5912–5931, 2006.
- [17] Lei Xu, Huiyong Sun, Youyong Li, Junmei Wang, and Tingjun Hou. Assessing the performance of mm/pbsa and mm/gbsa methods. 3. the impact of force fields and ligand charge models. *The journal of physical chemistry B*, 117(28):8408–8421, 2013.